

UTAS Social Network Analysis System

System Overview & Presentation Guide

Prepared 06 August 2026

A web-based Social Network Analysis platform that models student friendship networks at UTAS and turns that structure into concrete, evidence-based guidance for student political campaign planning — replacing guesswork about who to approach with an answer grounded in the actual shape of the student network.

1. What Problem Does It Solve?

Student political campaigns (TESCON and TEIN at UTAS) are normally planned on instinct: campaign organisers rely on personal familiarity and impressions of who is popular, with no structured evidence of how the student friendship network is actually shaped. That means campaign effort often goes to students who were already reachable anyway, while genuinely well-placed students, and the specific students who connect otherwise separate friend groups, get missed entirely.

This system fixes that by collecting real friendship data through a survey, modelling it as a graph, and running established social-network-analysis algorithms to surface exactly who matters and why.

2. How It Works — Data Flow

- **1. Data collection** — students complete a survey naming their friends in class, alongside college, department, year, and political-party affiliation.
- **2. Import** — the spreadsheet (CSV/Excel) is uploaded through the system; it is parsed and loaded into a relational database (students, connections).
- **3. Graph construction** — every student becomes a node, every friendship becomes an edge, using the NetworkX graph library.
- **4. Analysis** — the system computes centrality measures, detects communities, and identifies bridge nodes (see Section 3).
- **5. Presentation** — results appear on the Dashboard, the Analysis page, and an interactive Network Graph visualisation.
- **6. Action** — a one-click feature turns the analysis directly into a named, actionable campaign outreach plan (see Section 5).

Current live data in the system

Metric	Value
Total students	331
Total connections	197
Communities detected	140 (largest: 6 students)
High-influence students	33
Bridge nodes	0 (network is currently sparse/fragmented)

Note: the network is currently quite sparse (197 connections across 331 students), so most communities are small (pairs/triples). This is a genuine reflection of the imported survey data, not a system limitation — more complete survey responses would produce a denser, more connected picture.

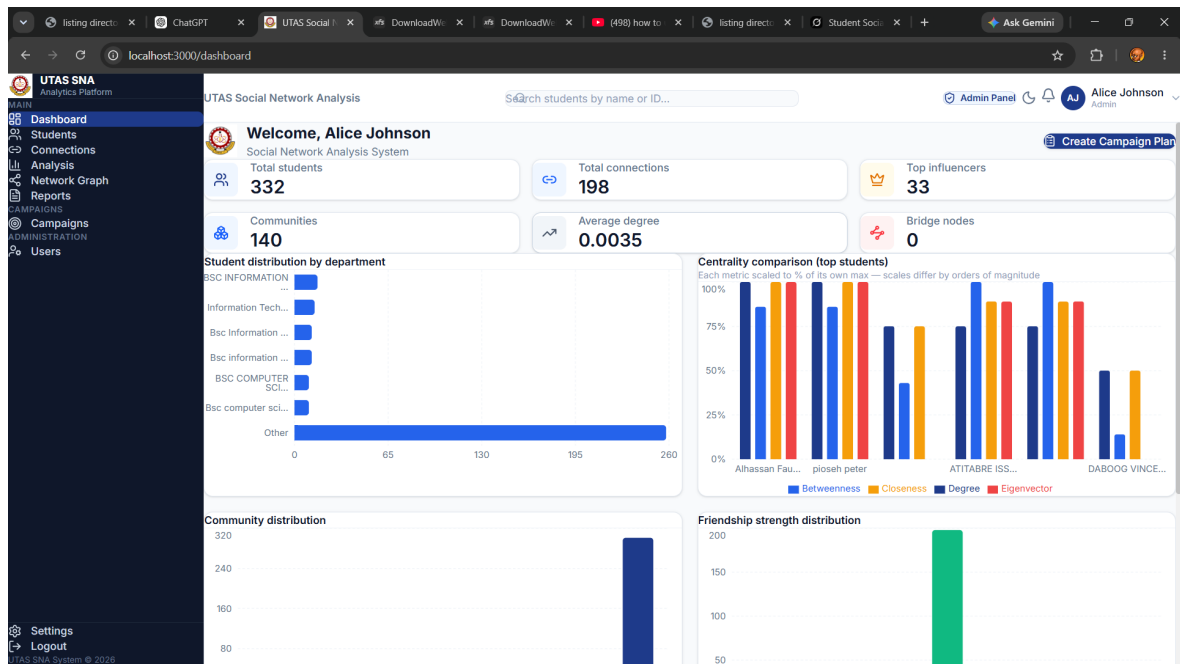
3. What the Analysis Actually Measures

Metric	What it tells you
Degree Centrality	How many direct friends a student has — simple popularity.
Betweenness Centrality	How often a student sits on the shortest path between two others — identifies brokers.
Closeness Centrality	How quickly a student can reach everyone else — identifies well-positioned students.
Eigenvector Centrality	Influence weighted by the importance of your friends, not just their number.
PageRank	Google's ranking algorithm applied to the friendship graph — a robust overall influence score.
Community Detection	Groups of students who are more tightly connected to each other than to the rest (Louvain method).
Bridge Nodes	The specific students whose removal would split the network into disconnected pieces.

4. Key Pages in the System

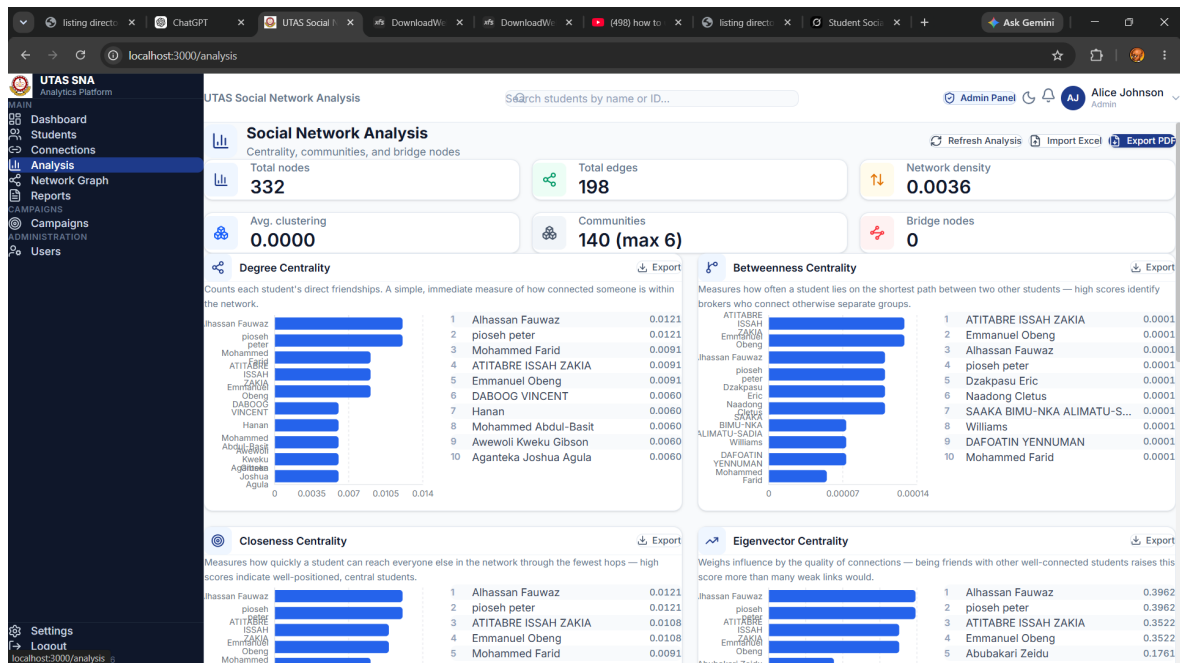
- **Dashboard** — KPI cards (students, connections, influencers, communities, bridge nodes) and charts.
- **Students / Connections** — searchable, filterable management tables for the underlying data.
- **Analysis** — a ranked table and chart for every centrality metric, with CSV export.
- **Network Graph** — an interactive, force-directed visualisation. Colour mode is a dropdown covering political party, detected community, department, tribe, religion, hometown, district, or regional capital, so any shared attribute can be seen as a cluster within the one graph; node size is driven by PageRank.
- **Campaigns** — create and track outreach campaigns, each with data-driven recommendations.
- **Reports** — generate PDF, CSV, and Excel exports of students, connections, and rankings.
- **Users / Settings** (Admin only) — account and role management, system information.

Dashboard



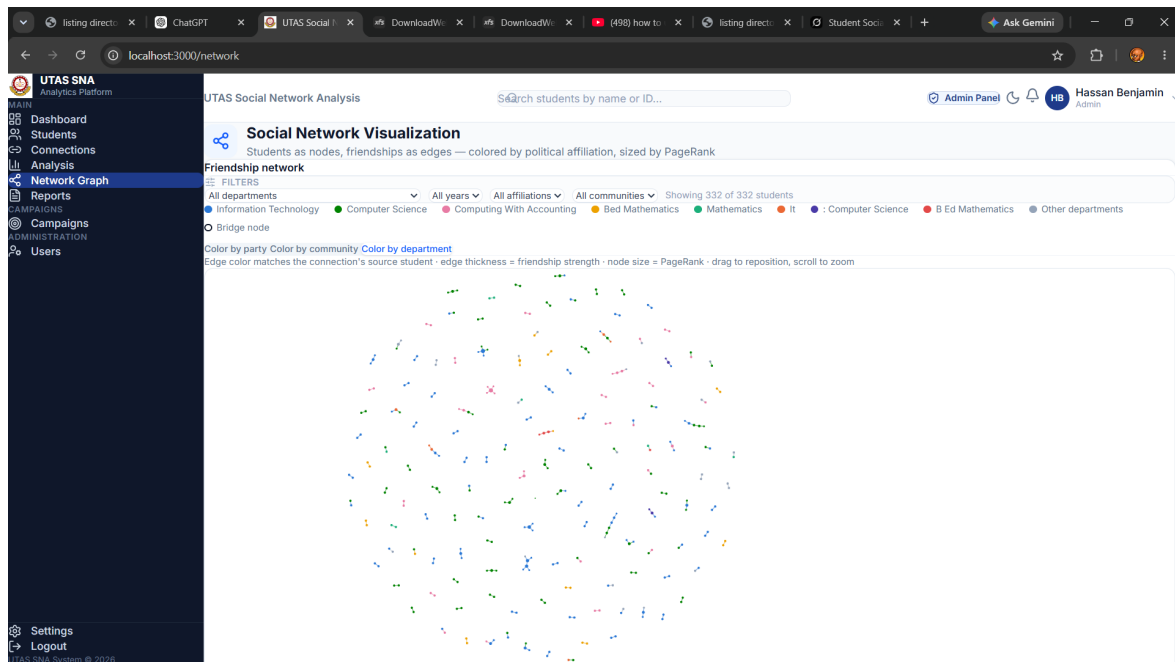
The Dashboard: KPIs, charts, quick actions, and recent activity.

Analysis



The Analysis page: ranked centrality metrics with descriptions and export.

Network Graph



The interactive Network Graph, coloured by department to show real cluster structure.

5. The Headline Feature: One-Click Campaign Planning

Clicking “Create Campaign Plan” on the Dashboard does not just open a form — it reads the live analysis and automatically creates a real campaign record with a concrete plan:

- **Target base** — which party's supporters and undecided students, sized against the current student count and number of communities.
- **Message amplification** — the top 3 influencers, named, to prioritise for outreach.
- **Cross-community reach** — the specific bridge students to engage so the message crosses clique boundaries instead of staying inside one group.
- **Coverage check** — flags communities with no high-influence student, so a volunteer can be assigned there directly.

This is the difference between a generic dashboard and a decision-support tool: the output is a specific, named action plan, not just a table of numbers.

6. Roles & Access Control

Role	Can do
Student (default)	View dashboard, analysis, network graph, and reports. Read-only.
Campaign Manager	Everything a Student can, plus: add/edit students & connections, import survey data, run analysis, plan campaigns.
Administrator	Everything a Campaign Manager can, plus: manage user accounts and view system information.

Every restriction is enforced on the backend API itself, not just hidden in the interface, so a Student account cannot reach a Campaign Manager's actions even by guessing a URL.

7. Live Deployment

The system is not just running locally — it is deployed and publicly reachable, on a genuinely free hosting setup (no billing account required anywhere):

Layer	Where it runs
Frontend	Firebase Hosting: myapplication-65e7d8b9.web.app
Backend API	PythonAnywhere: HBenzy.pythonanywhere.com
Database	SQLite, persisted on the backend host

This means the system can be opened and demonstrated from any device with a browser, right now, without needing a laptop running a local server — including live, during today's meeting.

8. Technology Stack

Layer	Technology
Backend	Python, Flask (REST API), SQLAlchemy (ORM), SQLite
Analysis engine	NetworkX (centrality, PageRank), python-louvain (community detection)
Authentication	JSON Web Tokens (JWT), role-based access control, Google Sign-In
Frontend	React, TypeScript, Tailwind CSS, React Router
Visualisation	Recharts (charts), react-force-graph-2d (network graph)
Reports	ReportLab (PDF), SheetJS/xlsx (Excel), native CSV export
Testing	pytest backend test suite (57 automated tests: auth, students, connections, campaigns, analysis, data import)
Hosting	Firebase Hosting (frontend), PythonAnywhere (backend)

9. Talking Points for Today's Meeting

If asked “what does this actually do differently from a spreadsheet?”

A spreadsheet shows who a student's friends are. This system computes structural properties of the whole network — who bridges groups, who has outsized influence, where the campaign's coverage has gaps — none of which is visible just by looking at raw rows of data.

If asked “is this your own original work?”

The system, its architecture, the algorithms' application to this specific problem, the campaign-planning feature, and the role-based access model are all built for this project. The underlying algorithms (NetworkX centrality measures, the Louvain method) are established, peer-reviewed techniques from the SNA literature, applied here to a new problem and dataset.

If asked “what are its limitations?”

Be upfront: the current dataset is sparse (197 connections for 331 students), so bridge-node and community results are limited by data completeness, not the method. The system does not yet do sentiment analysis or predictive modelling, both flagged as future work. A few individual survey responses had typos or data-entry mistakes that needed manual cleanup after import — normal for real, self-reported survey data, and now largely automated going forward.

If asked whether it is a real, working system or a prototype

It is live and publicly reachable right now (see Section 7) — not just running on a laptop. That covers the full path a real system needs: data import, a tested backend API, a role-based frontend, and actual deployment, not only the analysis code itself.

If asked to demo it live

Open it directly at myapplication-65e7d8b9.web.app on any device with internet access. Suggested order: Dashboard (overview) → Analysis (show the ranked metrics with descriptions) → Network Graph (switch the colour mode live — department, then tribe or hometown, to show the same students clustering differently by different attributes) → Dashboard again, click Create Campaign

Plan, then open the newly created campaign on the Campaigns page to show the generated plan naming real students.